PUBLIC HEALTH IMPACT · 1796 – 2020

The Power of Vaccination

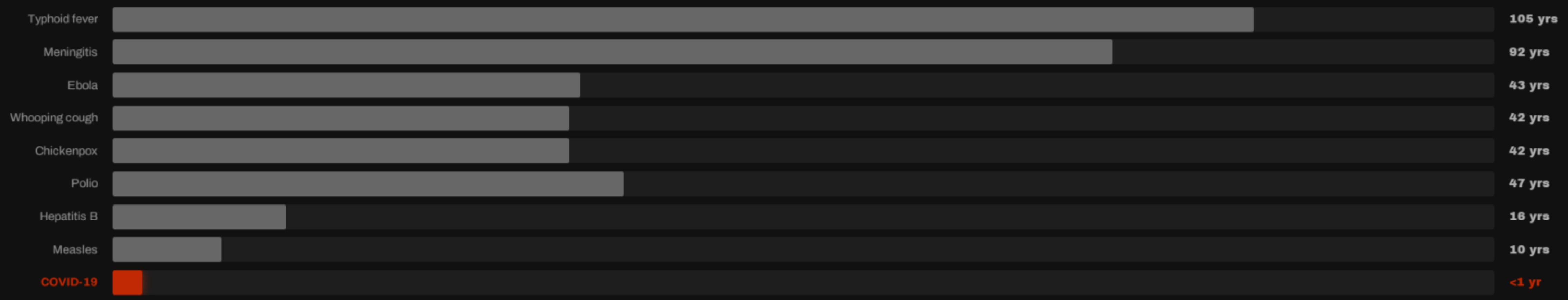
From Smallpox Eradication to COVID-19 in Record Time:
A data-driven overview of how vaccines have transformed public health over two centuries.

Source: Our World in Data · ourworldindata.org/vaccination

For public health professionals & policy analysts

DEVELOPMENT ACCELERATION

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



Years from pathogen identification to vaccine licensure. mRNA technology and genomic tools enabled the COVID-19 record. Source: Our World in Data.

UNITED STATES IMPACT

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

DISEASE	PRE-VACCINE CASES (PER MILLION/YR)	POST-VACCINE CASES (PER MILLION/YR)	CASE REDUCTION	DEATH REDUCTION
Diphtheria	158	0	100%	100%
Smallpox	250	0	100%	100%
Acute Polio	141	0	100%	100%
Measles	3,044	0.2	99.99%	100%
Rubella	242	0.04	99.98%	100%
Pertussis	1,534	52	96.6%	99.7%
Hepatitis A	465	51	89%	88.7%
Acute Hepatitis B	273	44	83.9%	83.6%
Pneumococcal	233	139	40.5%	31.3%

STATE-LEVEL IMPACT · 1929 – 2022

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Pre-Vaccine Era
1929 – 1962
100–1,000+ cases per 100,000
Nearly every state — high incidence

1963 Vaccine

Vaccine Era
1963 – 1979
Substantial decline across states
MMR introduced 1971

1980 Mandate

Post-Mandate Era
1980 – 2022
Near zero across all 50 states
Only sporadic outbreaks

1963

First measles vaccine developed by John Enders. Widespread adoption begins.

1971

MMR (measles-mumps-rubella) vaccine developed by Maurice Hilleman. Combined protection.

1980

Vaccination became mandatory for children entering kindergarten. Cases drop to near zero.

THE GREATEST VACCINATION SUCCESS

Smallpox: The Only Human Disease Eradicated Through Vaccination

1796 — **Jenner's First Vaccine**
Edward Jenner creates the first vaccine using cowpox to trigger immunity against smallpox.

1959 — **WHO Resolution**
World Health Organization passes resolution to eradicate smallpox globally.

1967 — **Intensified Campaign Launched**
WHO launches intensified eradication campaign with coordinated global surveillance and vaccination.

1977 — **Last Natural Case — Somalia**
The last naturally occurring case of smallpox recorded anywhere in the world.

1980 — **Eradication Certified**
WHO certifies global eradication — the only human disease eliminated through vaccination.

HUMAN-ONLY VIRUS
Smallpox infected only humans — no animal reservoir to sustain transmission.

VISIBLE SYMPTOMS
Distinctive rash made identification and containment straightforward.

EFFECTIVE & CHEAP VACCINE
Vaccine was stable, affordable, and highly effective across all populations.

COMMUNITY PROTECTION

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

Low vaccination coverage

Pathogen spreads freely — vulnerable individuals infected

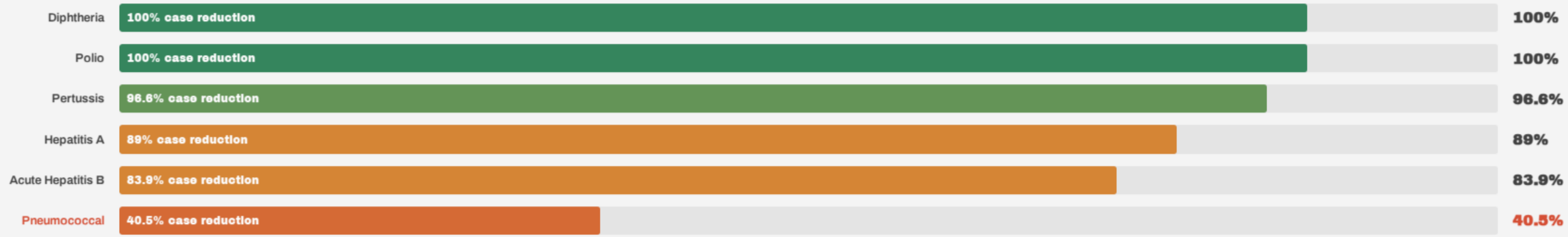
High vaccination coverage

Pathogen cannot spread — vulnerable individuals shielded

Vaccinated · Unvaccinated · Vulnerable (cannot be vaccinated: newborns, immunocompromised, chemo patients) · Infected · Vulnerable but protected (herd immunity)

WHERE WORK REMAINS

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction



Access Gaps

Globally, not everyone has access to vaccines. Supply shortages put children at risk in low-income regions.

Coverage Challenges

Incomplete immunization schedules and inequitable distribution limit herd immunity thresholds.

Trust & Uptake

Poor scientific understanding can reduce vaccine trust and uptake, allowing preventable resurgences.

SUMMARY & CALL TO ACTION

Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

01 **100% elimination achieved**
Vaccines achieved 100% case and death reduction for diphtheria, smallpox, and both forms of polio in the US.

02 **Development timelines collapsed**
From 105 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic tools.

03 **Measles near-eliminated in the US**
Cases dropped to near zero across all 50 states following the 1980 kindergarten vaccination mandate.

Improve access, coverage, and trust in vaccination — the tools exist; the commitment must follow.

CALL TO ACTION